

Different Sources of Energy

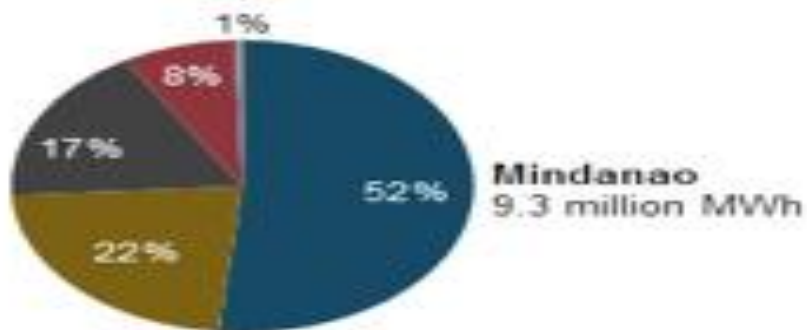
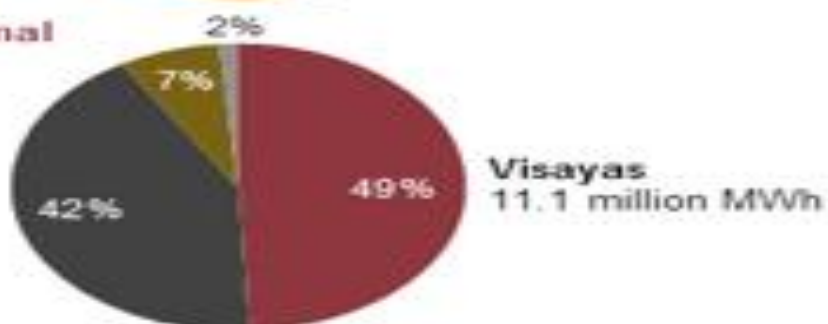
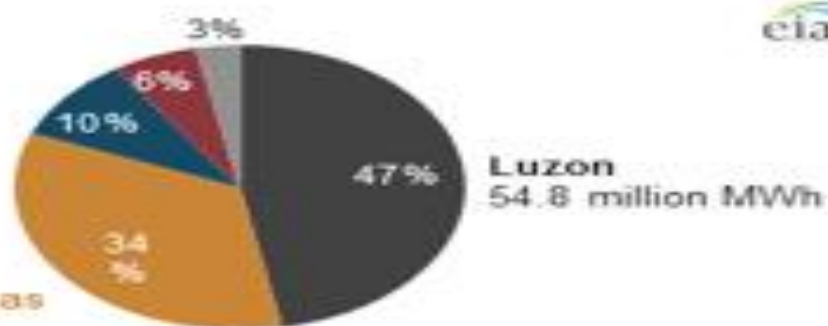


Gross power generation by fuel (2013)

Luzon
Visayas
Mindanao

Manila

coal
natural gas
oil
geothermal
hydro
other



ELECTRICITY AND WHERE IT COMES FROM

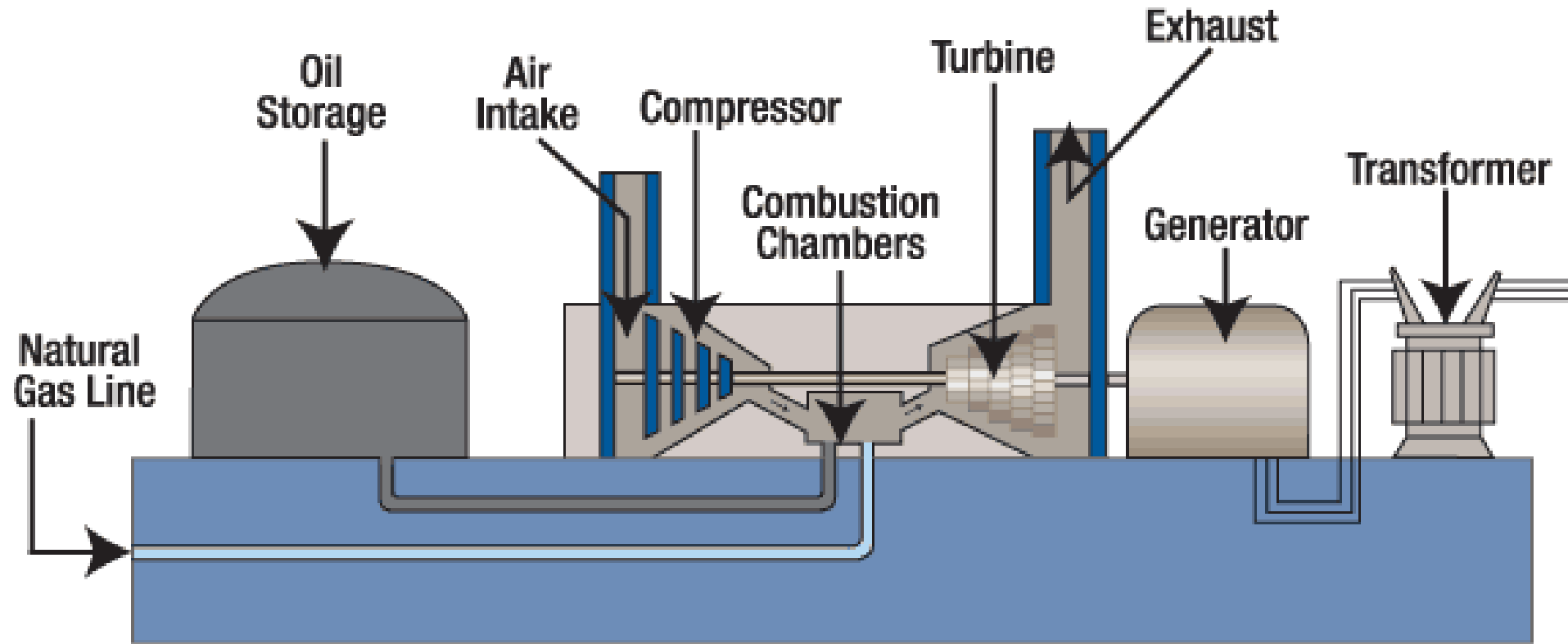
Sources of Energy can be broken down to two major groups: renewable and non-renewable sources.

Non-renewable	Renewable	
Fossil Fuel (Coal, petroleum, natural gas) Electrochemical (batteries)	Hydroelectric Solar power Hydrothermal	Wind power Geothermal Biomass

Non-renewable Energy Sources

Fossil Fuel

Fossil fuels, including coal, oil and natural gas, are currently the world's primary energy source.



Non-renewable Energy Sources

Fossil Fuel

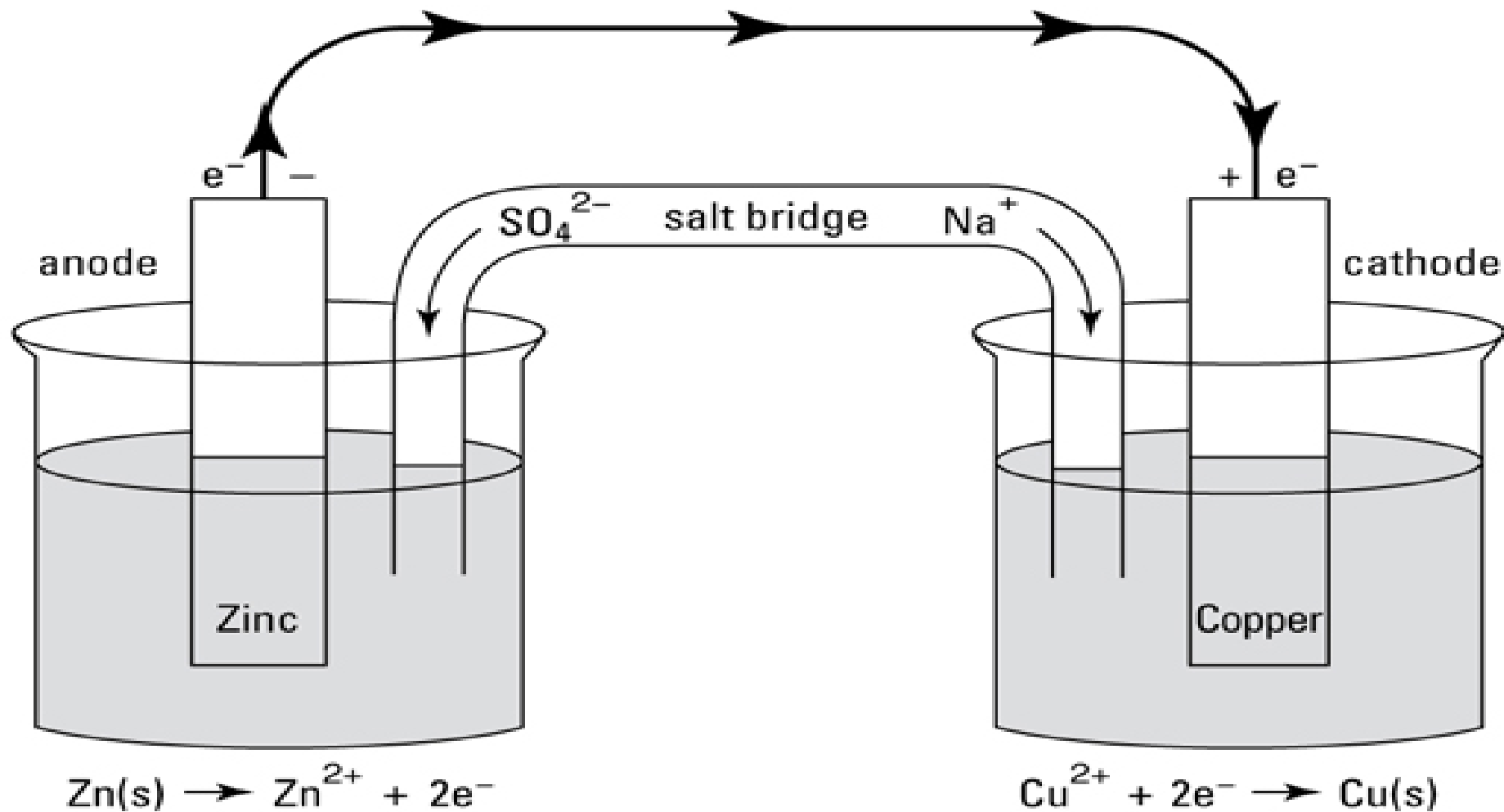
Fossil Fuels are burned to generate steam to power turbines and produce energy.

Fossil fuel Power Plants generate large amounts of carbon dioxide as a by product.

Non-renewable Energy Sources

Electrochemical Cells

Fuel cells that convert chemical energy into electricity through chemical reactions.



Non-renewable Energy Sources

Electrochemical Cells

The first battery was invented by **Alessandro Volta**, that propelled studies on electricity and allowed scientists to produce electricity.

Two main types of Battery:

Primary (Disposable Battery)

Alkaline, Mercury, silver-oxide and zinc carbon batteries

Secondary (Rechargeable Battery)

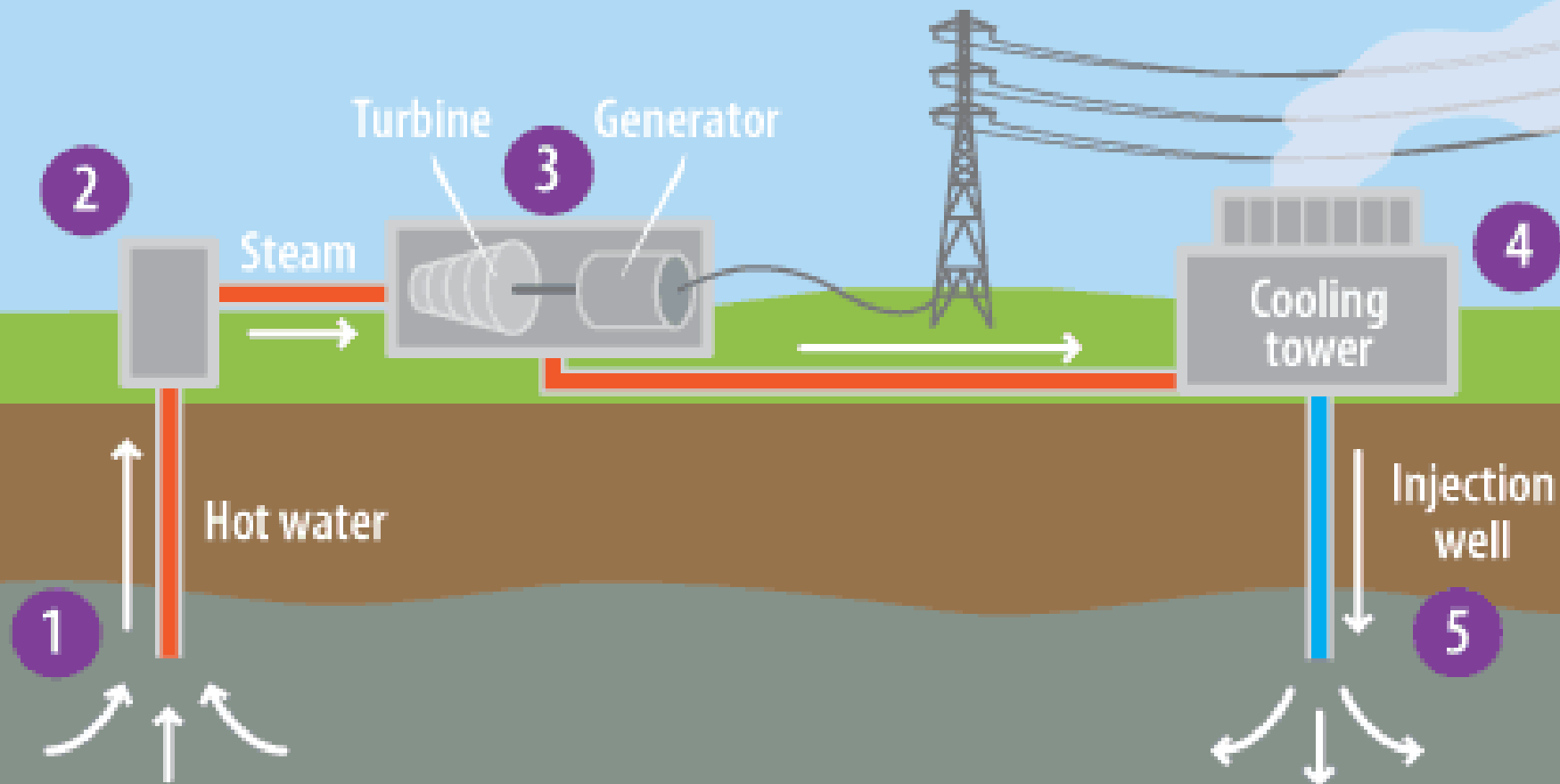
Lead-acid and lithium batteries

Renewable Sources of Energy

GEOHERMAL

Geothermal power utilizes the heat energy from the earth's crust.

Geothermal Power Plant

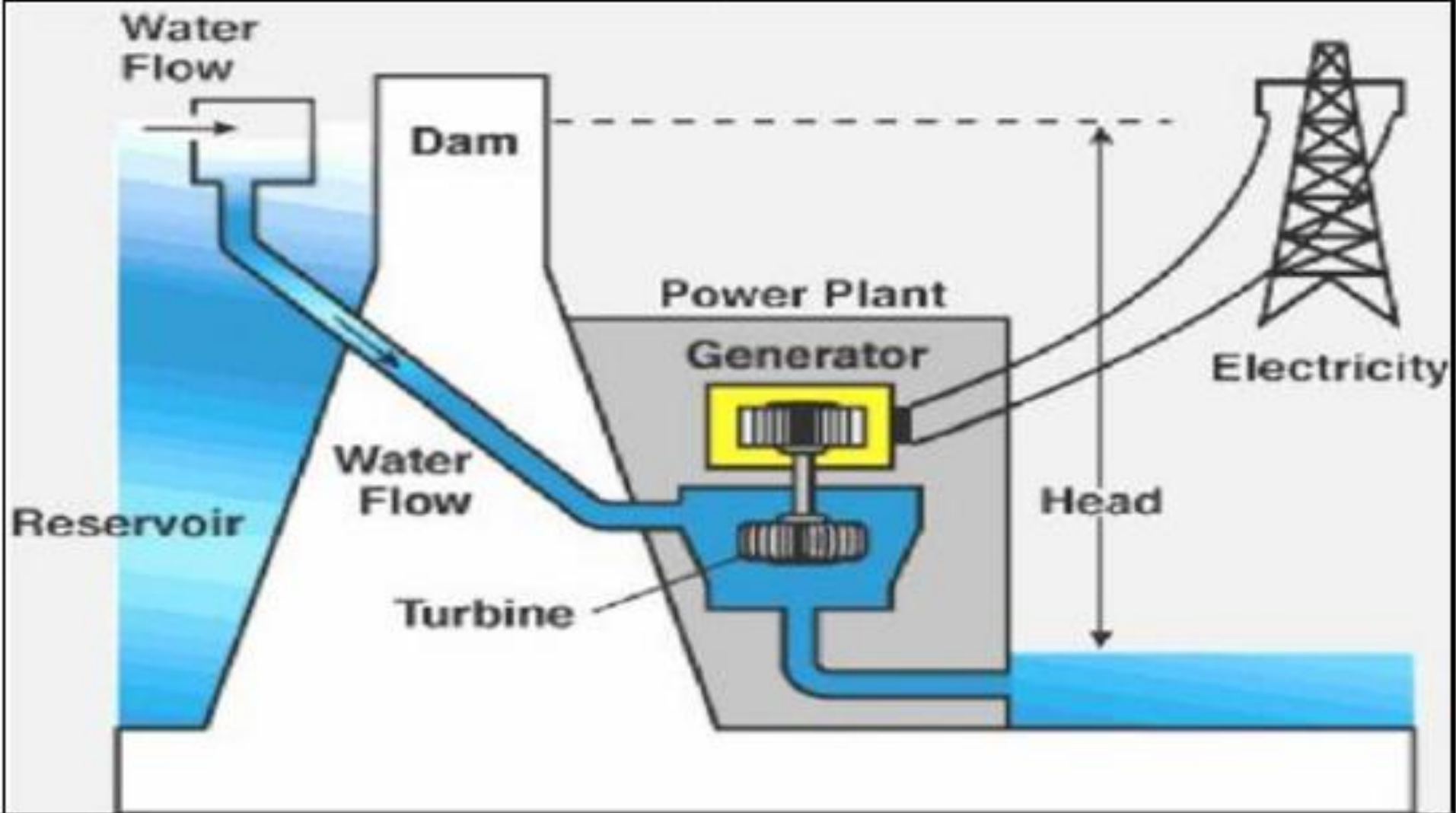


Renewable Sources of Energy

HYDROELECTRIC

Hydroelectric utilizes moving water to turn turbines.

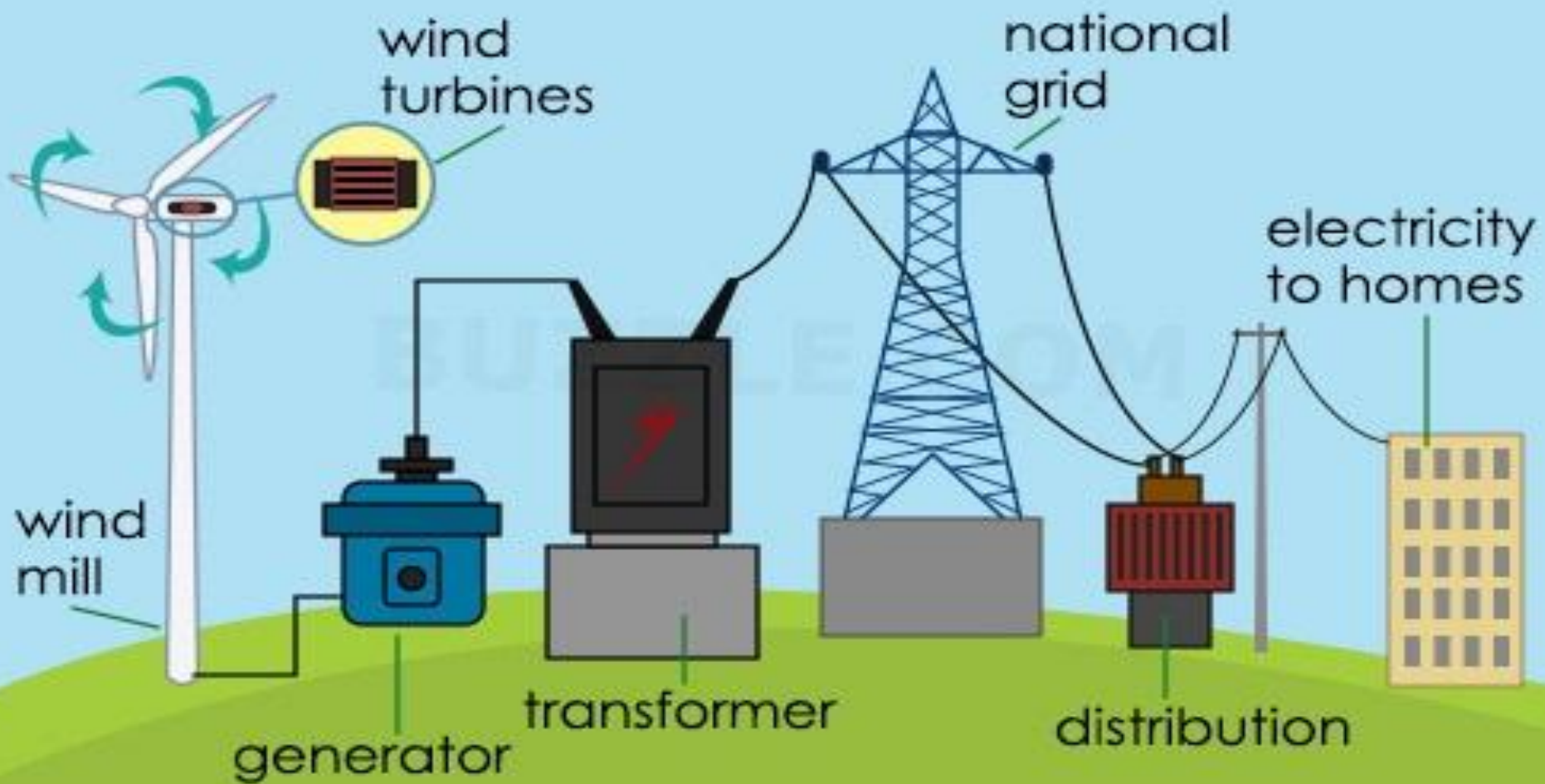
Dams are installed in strategic locations where there is a substantial drop in elevation.



Renewable Sources of Energy

WIND POWER

Wind turbines are installed in strategic locations to harness the mechanical energy from wind.



Renewable Sources of Energy

WIND POWER

Main examples of wind turbines in the Philippines are the ones found in San Lorenzo, Guimaras and Bangui Bay, Ilocos Norte

Renewable Sources of Energy

SOLAR CELLS

Solar panels are composed of photovoltaic cells or cells that convert light to electricity. When light shines on the panels, electrons are made to jump out of the free atom, thereby creating electricity.

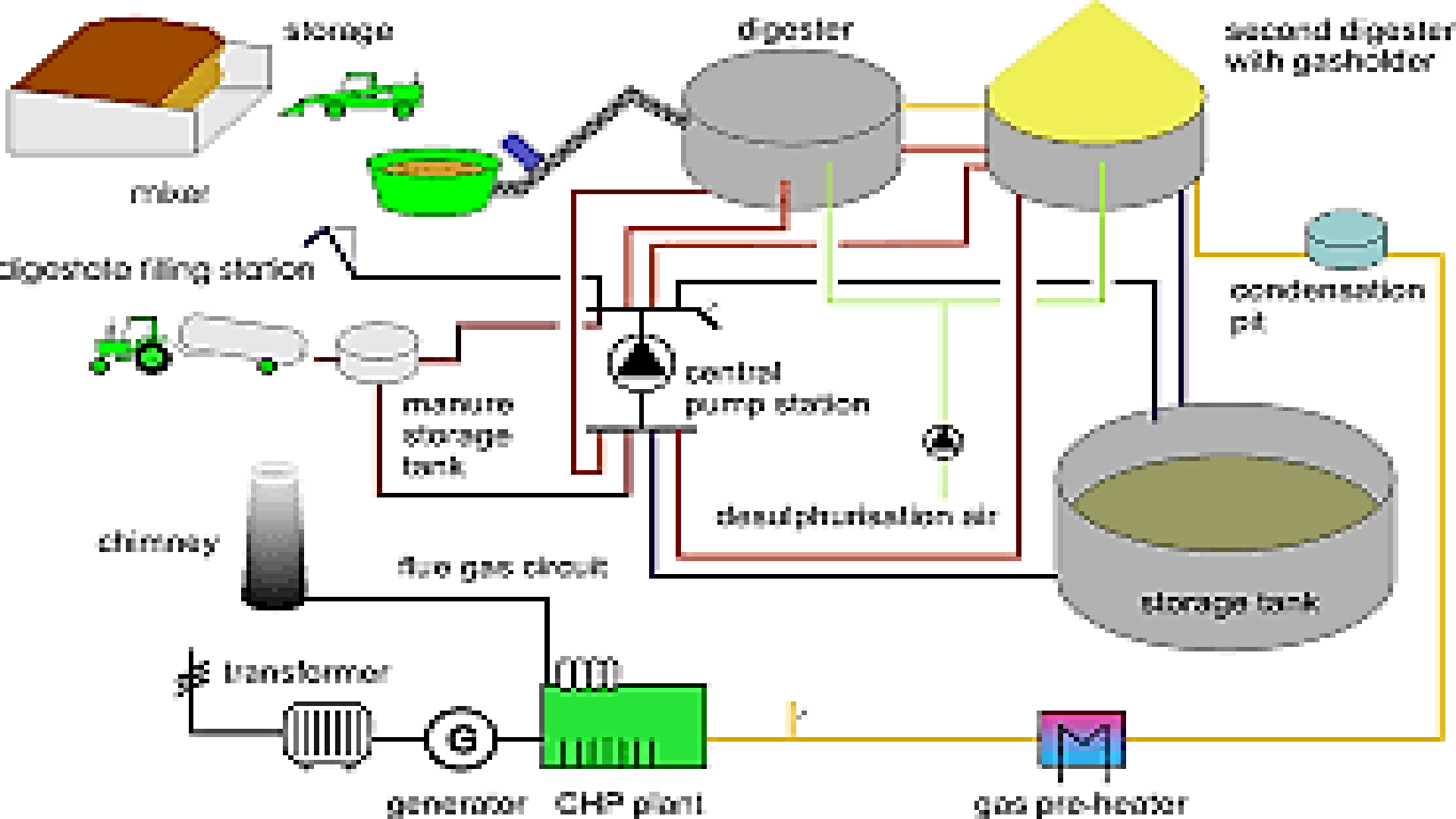


Renewable Sources of Energy

BIOMASS

Biomass refers to organic matter from plants and animals (microorganisms).

Wastes such as crop remains, manure and garbage are good sources of biomass.



Summary

Enumerate renewable and non-renewable sources of energy.

Describe where they extract energy.